

Model-level

H1: Composite orientation

$$\text{PVOC}_m \rightarrow \Delta_m$$

Aggregate PVOC index predicts cross-model burden effects

$N = 15$ model endpoints

$$\beta_1 = -2.14, 95\% \text{ CI } [-6.23, 1.95]$$

$$R^2 = 0.09$$

Null: no detectable association

Narrative-level

H2: Narrative frame association

$$\text{Frame}_{mi} \rightarrow \Delta_{mi}$$

Interpretive frames correlate with predicted burden effects

$N = 2,019$ rationale-effect observations

Access barriers $\uparrow$ ($b = 2.36$)

Co. responsibility n.s. ($b = 0.69$)

Coercive backlash $\uparrow$ ($b = 7.26$)

Partial support: 2 of 3 frames

Within-model

H3: Frame manipulation

$$\Delta_{mcir} \approx \text{Frame}_c$$

Assigned governance frame shifts predicted effect within the same model

$N = 4$ models $\times$ 27 profiles $\times$ 4 frames

Autonomy $\uparrow$ (+4.96), equity $\uparrow$ (+4.84)

Collective obligation $\downarrow$ (−10.88)

$$\text{adj. } R^2 = 0.77$$

Supported (pooled; model heterogeneity)

Does normative-frame sensitivity affect LLM-assisted policy simulation?

Three independent diagnostic tests — each informative regardless of the others.

PVOC null (H1) $\rightarrow$ frames partially matter (H2) $\rightarrow$ causal manipulation confirms (H3)